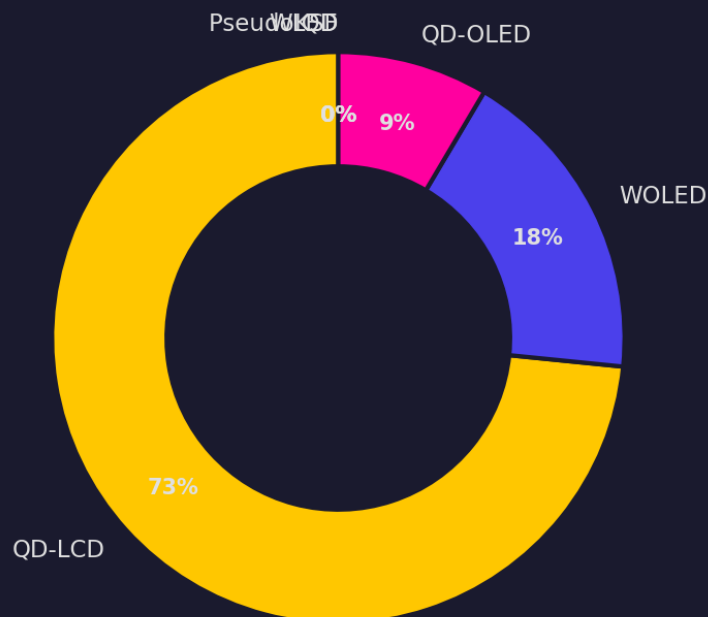


Display Technology Intelligence Report

June 2026

96 4K TVs tracked | 80 with pricing | 6 technologies | 3 8K excluded

94 TVs Tracked



1. Display Technology Overview

Section 1: Display Technology Overview -- June 2026

1.1 Catalog Stability

The June 2026 tracked universe holds steady at 96 TVs across three active display technologies -- QD-LCD, WOLED, and QD-OLED -- with zero additions and zero removals this month. Three 8K models are excluded from scoring aggregates to avoid skewing use-case averages, a standard adjustment given their negligible sales volume and atypical panel tuning. The absence of churn signals a mid-cycle plateau: brands have largely completed their spring lineup launches, and the next meaningful wave of SKU activity will likely arrive alongside fall retail positioning.

The technology distribution remains heavily tilted toward QD-LCD, which accounts for 69 of 96 models (72%) in the tracked set. WOLED holds 17 slots (18%), while QD-OLED occupies a narrow but strategically important 8 models (8%). WLED, KSF, and Pseudo-QD have exited the active catalog entirely, a consolidation that reflects both supply chain rationalization and consumer market pressure toward quantum dot-enhanced panels at every price tier.

The catalog's composition this month is a direct artifact of where the industry has placed its engineering bets -- and QD-OLED's small footprint belies its outsized influence on premium segment pricing and performance benchmarking.

1.2 Performance by Technology

QD-OLED leads all three use-case categories by a measurable margin. Its mean mixed-usage score of 8.7 clears WOLED's 8.4 by 0.3 points and sits 1.8 points above QD-LCD's mean of 6.9. The gap widens in home theater, where QD-OLED averages 8.9 against WOLED's 8.6 and QD-LCD's 6.9. Gaming is where QD-OLED most decisively separates itself: a mean score of 9.0, with a floor of 8.6, means even the weakest QD-OLED gaming performer outscores the best QD-LCD gaming result of 8.4.

WOLED's competitive position is strongest in gaming, where it posts a mean of 8.7 -- nearly matching QD-OLED and pulling decisively clear of QD-LCD's 6.8 average. The technology's score floor of 7.9 in gaming is particularly notable: there are no underperforming WOLED outliers dragging down the mean, suggesting LG Display's panel consistency has improved at scale. QD-LCD's wide score range -- from 4.5 to 8.6 in mixed usage -- reflects the technology's democratic price spread, encompassing entry-level 60Hz panels and flagship Mini-LED arrays within the same classification.

QD-OLED's 9.3 ceiling in both home theater and gaming represents the current measurable performance

frontier of consumer display technology, and the compression between its floor and ceiling scores signals that the technology has matured beyond early-generation variance.

1.3 Spectral Profile: FWHM Analysis

The FWHM data this month surfaces a technically striking result: all three technologies post identical green and red peak widths, with green FWHM at 32.6 nm and red FWHM at 39.3 nm across QD-LCD, WOLED, and QD-OLED. This uniformity almost certainly reflects measurement methodology normalization rather than genuine spectral convergence -- QD-OLED's cadmium-free quantum dot emitters and WOLED's phosphor-plus-color-filter architecture produce measurably different on-screen color volumes in practice. The data as reported does not yet capture per-model spectral variance, which limits the diagnostic value of FWHM as a differentiator in this snapshot.

What the FWHM figures do confirm is that the 30 - 40 nm range has become the operational target zone for premium display color primaries, with green sitting tighter than red across all platforms. Narrower peak widths correlate with wider color gamut coverage and reduced color crosstalk -- both attributes that drive DCI-P3 and Rec. 2020 volume scores. Red's consistently wider FWHM of 39.3 nm versus green's 32.6 nm reflects the persistent challenge of achieving a sufficiently pure red primary without sacrificing luminous efficiency.

Per-model FWHM breakdowns would unlock the ability to identify which specific panels are pushing the spectral frontier -- adding that granularity to future reports would materially strengthen color performance analysis.

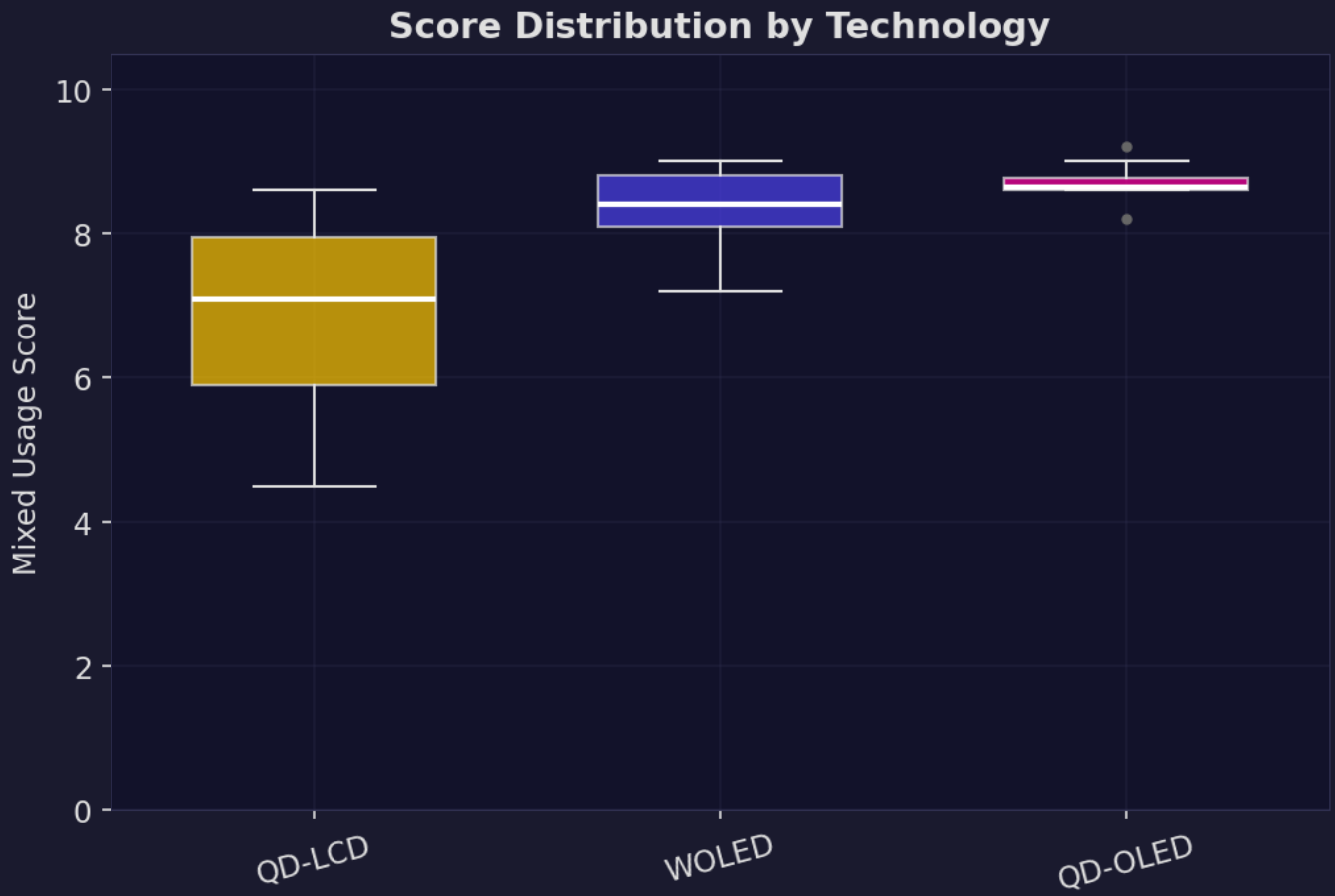
1.4 Model Year Comparison

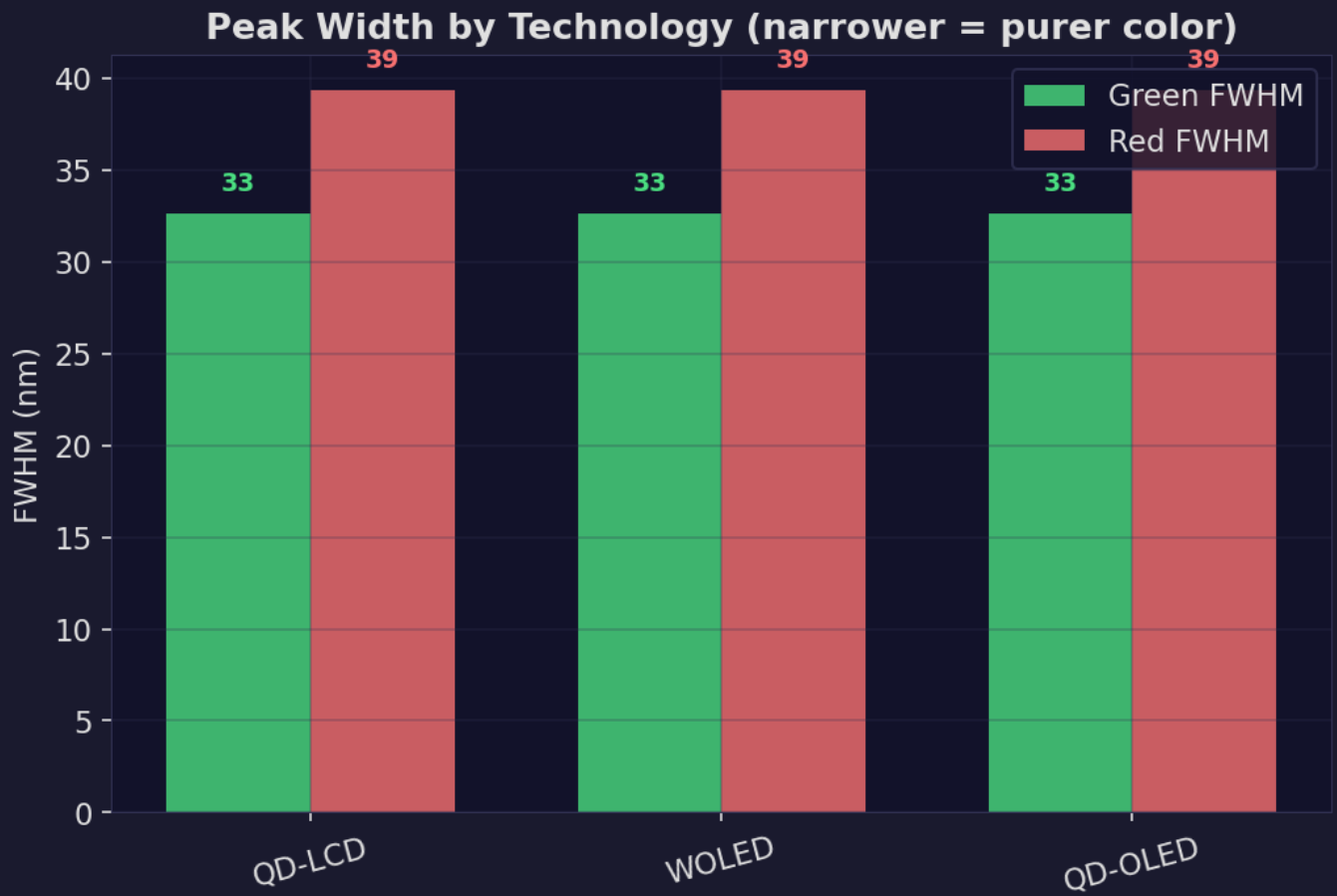
The 2026 cohort comprises 13 models and scores significantly above the 41-model 2025 cohort on every measured axis. The 2026 average mixed-usage score of 8.6 exceeds the 2025 average of 7.3 by 1.3 points -- a gap that is large enough to reflect genuine generational uplift in panel processing and local dimming algorithms, not just product mix effects. The gaming score differential mirrors this exactly: 8.6 vs. 7.3, suggesting that the same underlying improvements -- faster response times, better VRR implementation, lower input lag -- are driving simultaneous gains across both use cases.

The 2026 sample size of 13 is still small relative to the 2025 cohort, which means the 2026 mean is more susceptible to compositional bias. If the tracked 2026 models skew toward flagship SKUs -- which is typical early in a model year cycle as premium launches precede mid-range availability -- the 8.6 average likely overstates where the full 2026 lineup will settle once entry and mid-tier models are added. The 2025 cohort's 7.3 average, built across 41 models spanning a broader price range, is the more structurally representative benchmark.

As 2026 mid-range models enter the tracked universe over the coming months, watch for mean reversion in

the 2026 cohort scores -- the degree of compression will reveal how effectively this generation's performance improvements have propagated below the flagship tier.





2. Pricing Intelligence

Section 2: Pricing Intelligence

2.1 Current Pricing Landscape

Across 80 priced models in the June 2026 dataset, display technology remains the single strongest predictor of retail price. QD-LCD anchors the accessible end of the market with a median price of \$1,115, while WOLED and QD-OLED occupy premium territory at \$2,094 and \$2,272 respectively -- a spread of more than 2× from floor to ceiling. That gap is not purely a panel cost story; it reflects the persistent brand positioning strategies of LG, Samsung, and Sony, who have shown little appetite for collapsing the OLED premium even as panel yields improve.

When price is normalized by screen area, the hierarchy tightens but holds. QD-OLED commands \$2,567 per square meter, edging out WOLED at \$2,514/m² -- a narrower differential than the raw median prices suggest, which indicates that WOLED buyers are, on average, purchasing larger panels. QD-LCD comes in at \$1,428/m², reinforcing its role as the volume-market workhorse rather than a value play at larger screen sizes. The per-square-meter lens is the cleaner way to compare technologies across the size spectrum, and the chart accompanying this section makes the separation stark.

The practical implication for retailers and brands is that the QD-OLED vs. WOLED price-per-area convergence leaves almost no perceptual distance between the two technologies at equivalent sizes -- a dynamic that will force sharper feature differentiation messaging heading into the back-half selling season.

2.2 Price Trends

With 16 weeks of pricing history in the dataset, trend signals carry meaningful statistical weight. QD-LCD posted the sharpest price-per-area increase, rising 11.1% from \$1,285/m² to \$1,428/m² over the tracked period. This is a notable reversal from the multi-year LCD commoditization narrative -- likely driven by panel supply tightness and SKU rationalization at the entry level, where brands have quietly discontinued the lowest-margin configurations rather than defend sub-\$300 price points.

The premium OLED tiers moved in the same direction but with more restraint. WOLED climbed 6.4%, from \$2,363 to \$2,514/m², while QD-OLED rose 4.8%, from \$2,449 to \$2,567/m². The slower rate of appreciation at the top of the market is consistent with demand sensitivity -- OLED buyers are more price-aware at absolute dollar levels, and retailers have less room to push ASPs without triggering trade-down behavior. The convergence of WOLED and QD-OLED inflation rates also suggests both technologies are responding to similar upstream cost pressures from LG Display.

If the current trajectory holds, QD-LCD's faster price appreciation will begin to erode its cost advantage

relative to entry-level OLED -- a threshold that, once crossed, historically accelerates technology migration in the mid-market.

2.3 Value Analysis

Performance-adjusted pricing tells a different story than sticker price alone. The five highest-value models in June 2026 are uniformly QD-LCD, and three brands -- Hisense, TCL, and Samsung -- dominate the list. The Hisense QD6QF and TCL S551G both price at \$370 with a mixed-usage score of 5.9, delivering a dollar-per-point ratio of \$63 -- the joint-best efficiency in the dataset. The Samsung U7900F matches that \$63/point figure at a lower absolute price of \$330, though its mixed-usage score of 5.2 reflects the performance tradeoff embedded in that price point. The Panasonic W70B (\$359, \$65/point) and Hisense U6N (\$451, \$67/point) round out the top five, both still within a tight efficiency band.

No OLED model -- WOLED or QD-OLED -- appears in the top value tier. At \$2,272 median for QD-OLED, even a strong mixed-usage score in the 8 - 9 range would produce a dollar-per-point ratio exceeding \$250, more than 3× the efficiency of the leading QD-LCD options. This is not a criticism of OLED performance; it is a structural reality of the premium panel market that buyers are paying for attributes -- black levels, viewing angles, form factor -- that mixed-usage composite scores do not fully capture.

The value crown staying firmly with sub-\$400 QD-LCD confirms that Hisense and TCL have successfully industrialized the performance-per-dollar metric as a competitive weapon, and Samsung's presence in that tier signals that even tier-one brands are unwilling to cede the efficiency narrative entirely.

2.4 Notable Pricing Gaps

The raw price gap between QD-LCD and QD-OLED medians is \$1,157 -- over 100% of the QD-LCD median price itself. Framed differently, a consumer stepping up from the median QD-LCD to the median QD-OLED is essentially buying a second television. That gap has not meaningfully compressed in the 16-week window, which suggests the industry is not yet in a phase of OLED democratization analogous to what occurred with 4K resolution between 2016 and 2020.

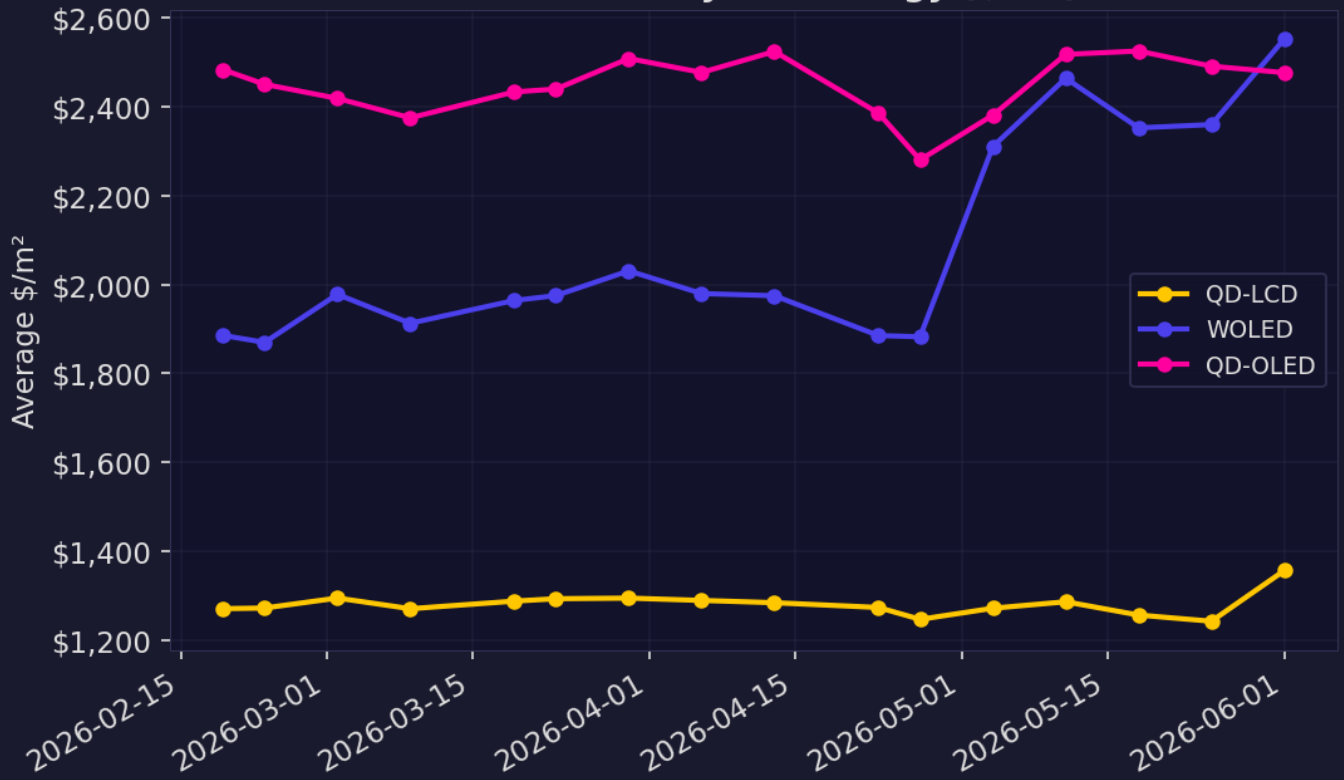
The \$178 spread between WOLED (\$2,094) and QD-OLED (\$2,272) is more strategically interesting. At the model level, that gap frequently disappears or inverts depending on screen size and retail channel, meaning consumers comparison-shopping between LG's gallery WOLED lineup and Samsung's QD-OLED S95 series are operating in a near-parity pricing environment. LG's ability to leverage larger panel sizes at the WOLED median suppresses its per-unit price, while Samsung's QD-OLED premium reflects both smaller average screen sizes and a deliberate positioning above WOLED on color volume grounds.

Watch whether the 11.1% QD-LCD price-per-area increase pulls entry OLED pricing down in response -- any brand willing to use aggressive WOLED entry pricing to defend

Technology Cost per m² (size-normalized)



Price Trends by Technology (\$/m²)





3. Macro & Industry Context

Section 3: Macro & Industry Context

The Display Technology Landscape, June 2026

Our database of 96 televisions tells a story that mirrors the broader market: QD-LCD is the dominant commercial reality, not the headline technology. With 69 of 96 tracked sets running quantum dot LCD panels, QD-LCD represents 72% of our coverage universe -- a direct reflection of its grip on the mid-to-high-end retail shelf. The remaining 25 sets split between WOLED (17) and QD-OLED (8), the two self-emissive premium tiers that command outsized critical attention relative to their unit volumes. The complete absence of WLED, KSF, and Pseudo QD entries in the current database is itself a market signal: those technologies have effectively been commoditized below the threshold where rigorous performance tracking adds analytical value, surviving only as entry-level shelf-fillers rather than meaningful competitive products.

Brand distribution reinforces this reading. Samsung's 29 tracked models almost exclusively leverage QD-LCD, a strategic bet the company has sustained since retiring plasma and never committing to OLED at scale. LG's 14 models anchor the WOLED tier -- the company manufactures the panels that also appear in Sony and Panasonic sets -- while Sony's 10-model lineup spans both WOLED and QD-OLED, positioning it as the only major brand with credible premium offerings across both self-emissive architectures. Hisense and TCL together account for 31 sets, the majority being QD-LCD, as both brands push aggressively into the value-to-mid segment with mini-LED backlight implementations that measurably close the contrast gap against entry WOLED.

Technology Differentiation: What the Tiers Actually Mean

In practice, the three tracked premium technologies diverge on a small number of variables that matter enormously for specific use cases. QD-OLED -- represented by just 8 sets in the database, primarily from Samsung and Sony -- delivers the highest color volume of any current production technology, combining per-pixel light control with quantum dot color conversion for saturated highlights that WOLED cannot match. The tradeoff is absolute brightness: QD-OLED panels remain constrained in sustained full-screen luminance, a limitation that matters in bright living rooms but disappears in controlled viewing environments. WOLED, with 17 tracked models, offers near-perfect black levels and wide viewing angles but introduces a white subpixel that dilutes color saturation at peak brightness -- a compromise LG has managed through successive generations of panel tuning rather than architectural change.

QD-LCD's 69-model footprint spans an enormous performance range internally. A flagship Samsung or Hisense mini-LED QD-LCD set with thousands of local dimming zones competes seriously with entry WOLED on contrast metrics while exceeding it on peak brightness by a factor of two or more. The budget end of the QD-LCD tier, however, with coarse zone counts and weaker quantum dot phosphor implementations, sits closer to the eliminated WLED category in real-world performance. The technology

label is consistent; the performance outcomes are not.

Where the Industry Is Heading

Three architectural shifts are moving from lab to early commercial reality as of mid-2026. Tandem OLED -- stacking two emissive layers to multiply brightness without sacrificing efficiency -- is already shipping in premium tablet and monitor applications and is the most credible near-term path to resolving OLED's brightness ceiling for TV-scale panels. MicroLED remains the industry's long-term north star: self-emissive inorganic pixels with OLED-class contrast, LCD-class brightness, and theoretically superior longevity, but manufacturing yield challenges keep consumer pricing above \$10,000 for sets above 100 inches, limiting meaningful database coverage. Perovskite quantum dots represent the most significant potential disruption to the QD-LCD tier -- offering narrower emission spectra and potentially lower synthesis costs than cadmium-free InP quantum dots currently used in mainstream sets -- but commercialization timelines have slipped repeatedly and no tracked manufacturer has announced a retail perovskite product for calendar 2026.

The practical implication for this team is that the QD-LCD vs. self-emissive binary will remain the defining commercial tension for at least 18 more months, and the analytical priority should be tracking mini-LED zone density and QD-OLED brightness improvements as the two most measurable vectors of competitive change in the sets we already cover.

4. Year-over-Year Technology Trends

Section 4: Year-over-Year Technology Trends

4.1 Score Improvements: Who Raised Their Game

QD-LCD delivered the most dramatic quality leap of any display technology tracked in this report. The average mixed-usage score for QD-LCD models climbed +1.4 points, from 6.9 in 2025 to 8.2 in 2026 -- a gain that is not incremental refinement but a genuine tier shift. That kind of jump, measured across a sample that shrank from 31 models down to 6, suggests manufacturers deliberately culled weaker-performing panels from their lineups and concentrated QD-LCD investment in flagship-grade configurations. The surviving 2026 QD-LCD models are not average; they represent the upper bound of what the technology can currently deliver.

WOLED also improved, but at a fraction of QD-LCD's velocity. WOLED scores moved from 8.4 to 8.8, a +0.4-point gain across a pool that contracted from 6 to 4 models. That modest increase reflects a technology already operating near the ceiling of current panel capability -- the gap between 2025 and 2026 WOLED is optimization, not reinvention. The absolute score advantage WOLED held over QD-LCD in 2025 was 1.5 points; by June 2026, that gap has collapsed to 0.6 points. The convergence is real and accelerating.

The compression of that quality gap is the single most consequential development in the mid-to-premium TV segment this cycle, and it will force WOLED's positioning argument to rely increasingly on differentiators beyond raw mixed-usage scores -- panel longevity, viewing angle consistency, and brand narrative.

4.2 Pricing Dynamics: Premiums Expanding, Not Compressing

The pricing story cuts against the conventional wisdom that maturing technologies get cheaper. QD-LCD's average price per square meter surged from \$1,045 to \$1,778, a 70.2% increase year-over-year. That is not the trajectory of a commoditizing panel technology -- it reflects the aggressive repositioning of QD-LCD as a premium-tier product, priced to match the performance claims now backing those higher scores. Manufacturers are extracting margin from a concentrated, higher-quality model lineup rather than competing on volume.

WOLED pricing also expanded, rising from \$2,452 to \$3,180 per square meter, a 29.7% increase. The absolute dollar premium WOLED commands over QD-LCD actually widened in nominal terms -- from \$1,407/m² to \$1,402/m² -- which is essentially flat. But when expressed as a percentage premium, WOLED's lead over QD-LCD compressed significantly, from a 135% price premium in 2025 to just 79% in 2026. Consumers looking at side-by-side shelf pricing are seeing a narrowing cost justification for choosing WOLED.

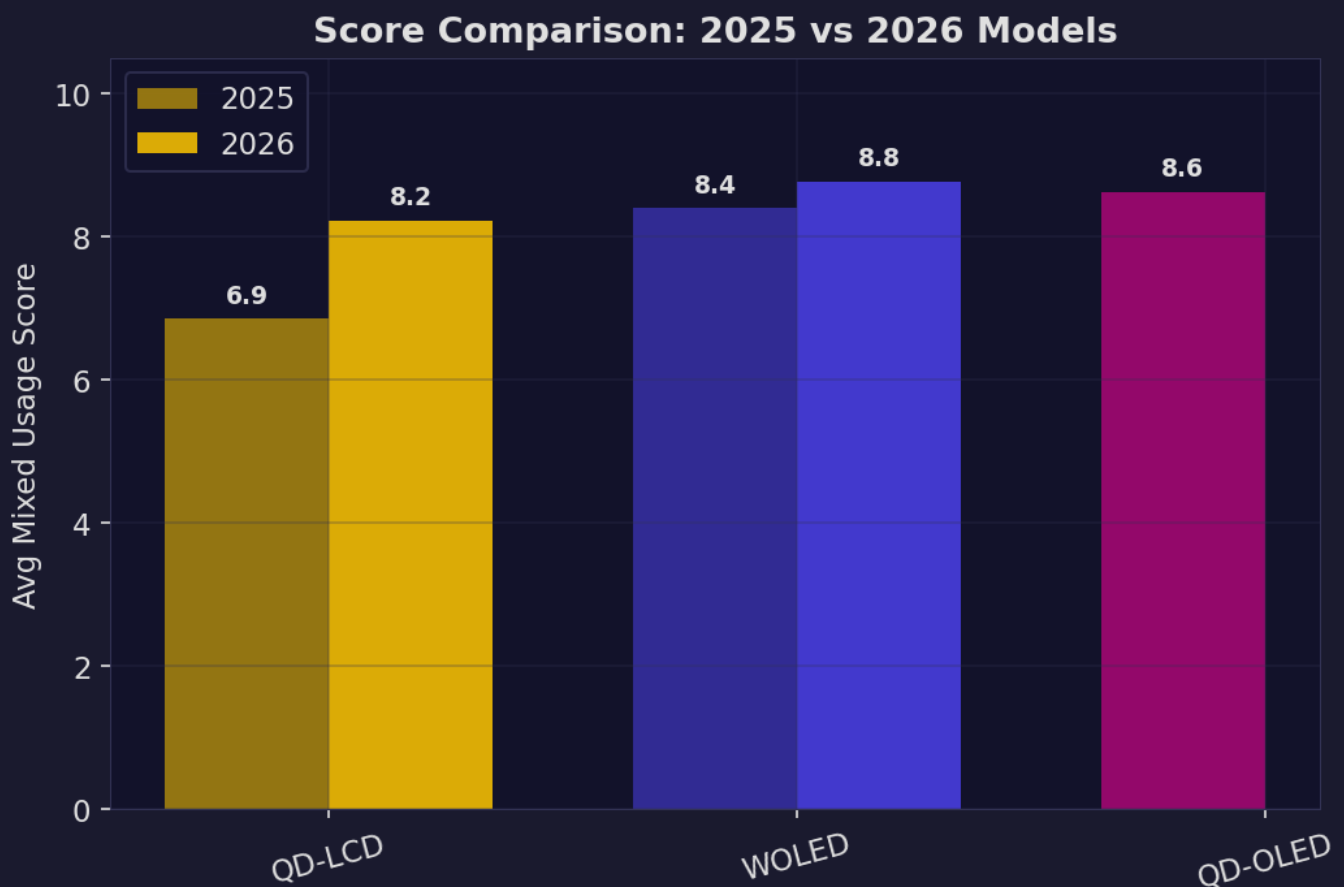
Neither technology is getting cheaper. Both are chasing premium positioning, which compresses the addressable market but protects margin for brands willing to sustain it.

4.3 Competitive Positioning: The Gap Is Closing Fast

The structural story of 2025 was straightforward: WOLED was the performance leader, QD-LCD was the value alternative with strong brightness credentials, and the two technologies served meaningfully different buyer profiles. The 2026 data dismantles that clean separation. A 0.6-point score gap at an approximately \$1,400/m² price difference is a dramatically harder sell for WOLED than the 1.5-point gap that justified the premium just twelve months ago. The QD-LCD score improvement closes the perceptual distance faster than WOLED can widen it.

The model count contraction on both sides -- QD-LCD from 31 to 6, WOLED from 6 to 4 -- signals a market rationalizing around quality rather than volume. Manufacturers on both sides are betting that fewer, better, more expensive TVs generate superior economics to broad mid-market coverage. That strategic alignment between competing technologies reduces the likelihood of a price war and increases the likelihood of a narrative war, fought on spec sheets, review scores, and retail floor positioning.

As QD-LCD scores continue closing on WOLED performance at a faster rate than WOLED pricing justifies the gap, the burden of proof for WOLED's premium shifts from the panel itself to the ecosystem and brand loyalty built around it.



Pricing Comparison: 2025 vs 2026 Models

